

Adelic-Irrational Mycelium Networks: Decentralized Bit-Level State Machines for Adaptive Bio-Inspired Edge Sensing and Computation

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1 Introduction

1.1 Problem Statement and Motivation

The contemporary development of intelligent systems and sensor interfaces is heavily dominated by centralized, resource-intensive approaches[cite: 9]. Whether deep learning or large-scale generative models, current paradigms demand substantial computing power, continuous cloud connectivity, and a rigid, pre-trained parameterization[cite: 9]. For applications in Edge-AI, decentralized environmental sensing, or soft robotics, these classical architectures quickly hit physical and energetic walls[cite: 9].

Nature resolves these challenges through entirely different design principles[cite: 9]. Biological organisms, such as fungal mycelia or decentralized ganglionic networks, operate without a centralized processing unit[cite: 9]. Instead, they process stimuli through local, non-linear interactions, adapting organically to their environment over time[cite: 9].

1.2 The Paradigm of the Mycelium Network

This paper introduces a novel biomimetic architecture: the Mycelium Network[cite: 9]. Rather than relying on heavy neural weights or deterministic execution loops, we implement a decentralized topology consisting of 17 interconnected processing cores[cite: 9]. Each core acts as an autonomous cellular unit that exchanges states via basic bitwise operations (XOR and bit-shifts)[cite: 9]. This topology mathematically mimics the moisture- and nutrient-based signal propagation characteristic of biological fungal networks[cite: 9].

1.3 The Mathematical Core: Adelic Structures, Primes, and Irrational Weights

To prevent artificial stagnation and repetitive cycles while ensuring multi-scale robustness, the architecture incorporates advanced principles from higher number theory[cite: 9]:

- **The Adelic Framework and p -Adic Base (17):** The system state is embedded into an adèle ring structure, using the prime number 17 as an essential anchor for discrete, coprime bit diffusion at the micro-level[cite: 9].
- **Irrational Constants (ϕ, π, e , etc.):** Each of the 17 cores is weighted with a fundamental irrational constant[cite: 9]. This enforces an aperiodic, ergodic exploration of the state space, completely banishing mechanical sterility[cite: 9].
- **Epigenetic Persistence:** Through local data serialization (`mycelium_dna.dat`), the system maintains a historical “memory,” allowing the network to continuously evolve and align with the specific stimulus patterns of its user over time[cite: 9].

1.4 Objective and Scope

In the subsequent sections, we formalize the mathematical dynamics of the irrational 17-core network through the lens of adelic systems, analyze its stability under real-time boundary conditions, and discuss concrete application scenarios for self-growing, adaptive bio-sensors of the next generation[cite: 9].

2 Mathematical Formulation and the Adelic System Model

2.1 Decentralized Core Topology and Register Architecture

The Mycelium Network is modeled as a decentralized system comprising $N = 17$ processing nodes (cores), denoted as $C = \{c_0, c_1, \dots, c_{N-1}\}$ [cite: 9]. Each core c_i maintains an independent 64-bit integer register $r_i \in \mathbb{U}_{64}$ representing its local state analogous to a biological fungal hypha[cite: 9]. The network topology combines a closed ring structure with local interactions to facilitate decentralized signal diffusion[cite: 9].

2.2 The Adelic State Vector ($Z_{n,\text{adel}}$)

To rigorously formalize the multi-scale dynamics of the network, the total system state at a discrete step n is embedded into the adelic ring $\mathbb{A}$ as an adelic state vector $Z_{n,\text{adel}}$ [cite: 9]. This formulation unifies continuous macroscopic behavior with discrete prime-channel representations[cite: 9]:

$$Z_{n,\text{adel}} = \begin{pmatrix} Z_{n,\infty} \\ Z_{n,p_1} \\ Z_{n,p_2} \\ \vdots \\ Z_{n,p_\tau} \end{pmatrix} \in \mathbb{A}$$

The components fulfill distinct functional roles within the system model[cite: 9]:

- **The continuous (archimedean) phase space vector** ($Z_{n,\infty} \in \mathbb{C}$): Represents the global, macroscopic dynamics of the network[cite: 9].
- **The discrete (p -adic / non-archimedean) channel vectors** ($Z_{n,p_i} \in \mathbb{Q}_{p_i}$): Form isolated information channels dedicated to individual prime numbers $p_1, p_2, \dots, p_\tau$, where the foundational prime base $p_1 = 17$ anchors the primary core interactions[cite: 9].

2.3 Irrational Weighting, Phase Angles, and Subfunctions

To avoid deterministic stagnation and short periodic loops common in conventional digital automata, each core c_i is permanently bound to a unique fundamental irrational constant $w_i \in W$ [cite: 9]. These weights are scaled integer representations of constants such as the golden ratio (ϕ), the circle constant (π), Euler's number (e), and chaos-theoretical values[cite: 9]:

$$W = \{w_0, w_1, \dots, w_{16}\} \subset \mathbb{Z}$$

The internal temporal evolution and coupling across channels are governed by precise deterministic subfunctions and parameters[cite: 9]:

- **Phase angle** (θ_n): Defined as $\theta_n = w \cdot n$ [cite: 9].
- **Resonance term** ($R_{n,i}$): Each prime channel p_i maintains an independent resonance progression[cite: 9]:

$$R_{n,i} = \sin(p_i \cdot n + w \cdot \sqrt{p_i} + W_{n,i})$$

- **Wave modulation** ($W_{n,i}$): Controls non-linear internal fluctuations via[cite: 9]:

$$W_{n,i} = A_i \cdot \cos(m_i \cdot \theta_n + \phi_i)$$

- **p -adic phase mapping** ($\chi_{p_i}(\theta_n)$): A transformation mapping $\mathbb{C} \rightarrow e^{i\theta_n} \rightarrow \mathbb{Q}_{p_i}$ that translates continuous Euler rotation $e^{i\theta_n}$ into the respective p_i -adic number field (e.g., via Teichmüller lift or p -adic characters)[cite: 9].

2.4 Uncorrupted Adelic Master Iteration and Diffusion Dynamics

When an external stimulus $S = (p, e)$ —consisting of a digitized pitch p and normalized energy e —is captured by the sensor interface, it impacts the root node c_0 [cite: 9]. The global temporal evolution of the entire system across both continuous Archimedean and discrete non-Archimedean p -adic prime channels from step n to $n + 1$ is governed conclusively by the uncorrupted adelic master iteration function[cite: 9]:

$$Z_{n+1,\text{adel}} = \begin{pmatrix} Z_{n+1,\infty} \\ Z_{n+1,p_1} \\ Z_{n+1,p_2} \\ \vdots \\ Z_{n+1,p_\tau} \end{pmatrix} = \begin{pmatrix} \left(Z_{n,\infty}^{\frac{1}{b \cdot w}} + \sum_{i=1}^{\tau} \frac{g_i R_{n,i}}{p_i - \tau} \right) \cdot e^{i\theta_n} \\ (s \cdot w) \cdot Z_{n,p_1} + \frac{g_1 R_{n,1}}{p_1 - \tau} \cdot \chi_{p_1}(\theta_n) \\ (s \cdot w) \cdot Z_{n,p_2} + \frac{g_2 R_{n,2}}{p_2 - \tau} \cdot \chi_{p_2}(\theta_n) \\ \vdots \\ (s \cdot w) \cdot Z_{n,p_\tau} + \frac{g_\tau R_{n,\tau}}{p_\tau - \tau} \cdot \chi_{p_\tau}(\theta_n) \end{pmatrix}$$

The complex power $Z_{n,\infty}^{\frac{1}{b \cdot w}}$ is evaluated strictly on the principal branch, defined via $Z^p = \exp(p \cdot \text{Log}(Z))$, where $\text{Log}(Z) = \ln |Z| + i \text{Arg}(Z)$ with principal argument $\text{Arg}(Z) \in (-\pi, \pi]$ [cite: 9].

At the hardware register level, the local update equation for the root core incorporates the quantized energy stimulus $\tilde{e} = \text{clamp}(e \cdot 200, 1, 16)$ [cite: 9]:

$$r_0^{(t+1)} = (r_0^{(t)} \cdot w_0) + (p \bmod 17) + \tilde{e}$$

For all subsequent cores c_i ($i \in \{1, \dots, 16\}$), the signal propagates through a non-linear bit-mixing function combining bitwise XOR operations, directional bit-shifts, and individual irrational scaling within the ring topology[cite: 9]:

$$r_i^{(t+1)} = \left(\left(r_i^{(t)} \oplus (r_{\text{prev}}^{(t)} \gg 3) \oplus (r_{\text{next}}^{(t)} \ll 5) \right) \cdot w_i \right) + (p \bmod 17)$$

where $\text{prev} = (i - 1) \bmod 17$ and $\text{next} = (i + 1) \bmod 17$ denote spatial neighbors[cite: 9].

2.5 Epigenetic State Persistence and System Phenomenology

To mimic long-term biological adaptation without heavy parameter optimization, the global network state $R^{(t)}$ is derived via a folded bitwise operation across all registers[cite: 9]:

$$R^{(t)} = \bigoplus_{i=0}^{16} r_i^{(t)}$$

Upon system termination, the vector of all registers $R_{\text{dna}} = (r_0, r_1, \dots, r_{16})$ is serialized into a persistent storage file (`mycelium_dna.dat`)[cite: 9]. This mechanism establishes an epigenetic memory, allowing the network to retain historical interaction patterns across operational sessions[cite: 9].

The structural interplay of these mathematical mechanics yields two foundational phenomenological properties within the model[cite: 9]:

- **Perfect symmetry in continuous space ($Z_{n,\infty}$):** All prime-number channels superpose harmonically to generate an ideal, interference-free 3D holographic projection of the system state[cite: 9].
- **Isolated information carriers (Z_{n,p_i}):** Each prime channel p_i processes exclusively its designated resonance term $R_{n,i}$ within its respective p_i -adic subspace, preventing cross-talk and data corruption[cite: 9].

3 System Architecture and Implementation

3.1 Computational Model and Register Mechanics

The practical realization of the Adelic-Irrational Mycelium Network is designed for high-efficiency, ultra-low-power execution environments, bridging theoretical number theory with low-level bit manipulations[cite: 9]. The architecture avoids floating-point arithmetic overhead in its core propagation loop by operating entirely on primitive 64-bit unsigned integer registers ($\mathbb{U}_{64}$)[cite: 9].

The 17 processing nodes are instantiated as a contiguous memory vector, ensuring optimal cache locality and deterministic instruction pipelining[cite: 9]. Each core executes independent bitwise transformations concurrently or sequentially within a tightly bounded execution cycle, mimicking the parallel chemical signaling pathways observed in biological hyphal networks[cite: 9].

3.2 Lightweight Signal Acquisition via Average Magnitude Difference Function (AMDF)

To process real-time environmental stimuli (such as acoustic waves or sensor transducers) without resorting to computationally expensive Fast Fourier Transforms (FFT), the system implements a streamlined Average Magnitude Difference Function (AMDF)[cite: 9]. The incoming signal buffer history of size N_h is evaluated across a defined lag interval $[L_{\min}, L_{\max}]$ [cite: 9]:

$$\text{AMDF}(\text{lag}) = \frac{1}{N_{\text{eval}}} \sum_{j=0}^{N_{\text{eval}}-1} |\text{history}[N_h - 1 - j] - \text{history}[N_h - 1 - j - \text{lag}]|$$

This time-domain periodicity detection isolates fundamental frequencies with minimal CPU cycles[cite: 9]. The estimated frequency is subsequently digitized and mapped into the discrete prime domain ($p \bmod 17$), serving as the primary biological trigger that disrupts or accelerates the internal diffusion dynamics of root node c_0 [cite: 9].

3.3 Non-Linear Feedback and Filter Mapping

The global combined state of the network, derived from the folded XOR operation $R^{(t)} = \bigoplus_{i=0}^{16} r_i^{(t)}$, acts as a multi-channel control voltage (CV) generator[cite: 9]. Bit segments of $R^{(t)}$ are sliced dynamically to modulate synthesis parameters and filter coefficients[cite: 9]:

- **Pitch and Scale Quantization:** Lower bits ($b_0 - b_3$) select discrete frequency steps from non-linear microtonal scales (e.g., cybernetic or industrial scaling arrays)[cite: 9].
- **Resonance and Cutoff Modulation:** Intermediate bit slices govern dual-stage filter cascades (modeled after analog MS-20 filter dynamics), regulating high-pass and low-pass cutoff frequencies alongside feedback resonance without relying on external control hardware[cite: 9].

4 Formal Homomorphism: From p-adic Topology to Native Hardware Level

A central challenge in implementing the adelic master iteration $Z_{n,\text{adel}} \in \mathbb{A}$ on conventional microprocessors lies in the translation of continuous and p-adic metrics into discrete register operations. In this section, we define the explicit projection morphism that proves the hardware-level integer arithmetic is an exact topological equivalent of the 2-adic number space $\mathbb{Q}_2$.

4.1 Truncation of $\mathbb{Q}_2$ onto the 64-Bit Register

An ordinary 64-bit CPU hardware register deterministically computes modulo 2^{64} upon overflow. Mathematically speaking, the set of unsigned 64-bit integers $\mathbb{U}_{64}$ thus does not form a Euclidean space, but rather the residue class ring $\mathbb{Z}/2^{64}\mathbb{Z}$.

Since the ring of 2-adic integers $\mathbb{Z}_2 \subset \mathbb{Q}_2$ is defined as the inverse limit $\varprojlim \mathbb{Z}/2^k\mathbb{Z}$, each 64-bit register represents exactly the first 64 terms of an infinite 2-adic expansion. The mapping of the theoretical p-adic sequencer channels Z_{n,p_i} onto the hardware is consequently not a heuristic hack, but a strict homomorphism Φ :

$$\Phi : \mathbb{Z}_2 \rightarrow \mathbb{Z}/2^{64}\mathbb{Z}, \quad x \mapsto x \bmod 2^{64} \quad (1)$$

4.2 Hensel Division as a Logical Bit-Shift

The deterministic core of the state update in the 2-adic space is the Hensel division. Let the state be given by the fraction $Z_{n,p} = \frac{a_n}{b}$, where the denominator b is odd. The extraction of the next opcode $b_n \in \{0,1\}$ and the advancement of the state vector ($n \rightarrow n+1$) are theoretically defined as:

$$b_n = |a_n| \pmod{2} \quad (2)$$

$$a_{n+1} = \frac{a_n - b_n \cdot b}{2} \quad (3)$$

Under the projection into the ring $\mathbb{Z}/2^{64}\mathbb{Z}$, the operation $b_n = |a_n| \pmod{2}$ corresponds exactly to the logical AND operator (`a_n & 1`) at the bit level. Since the subtraction ($a_n - b_n \cdot b$) guarantees that the least significant bit (LSB) becomes zero, the subsequent division by 2 corresponds to an exact arithmetic right shift. The mathematically complex state update collapses at the hardware level to an atomic $O(1)$ operation:

$$a_{n+1} \equiv (a_n - (a_n \& 1) \cdot b) \gg 1 \pmod{2^{64}} \quad (4)$$

This proof demonstrates that bit-shifts in the CPU topologically correctly navigate the ultrametric distance in the 2-adic tree.

4.3 The Archimedean Space $\mathbb{C}$ and CORDIC Projection

While the p-adic space natively fits into integer registers, the continuous phase space $Z_{n,\infty} \in \mathbb{C}$ typically requires a floating-point unit (FPU) for the Euler rotations $e^{i\theta_n}$. To maintain the paradigm of FPU-free, stateless system iteration, we transform the continuous space into a Q32.32 fixed-point format.

By applying the CORDIC (Coordinate Rotation Digital Computer) algorithm, all complex rotation terms and saturation curves (such as $\tanh(K \cdot x_n)$) are decomposed into pure additions and shift operations ($\gg$). The adelic master equation:

$$Z_{n+1,\infty} = \left((s \cdot w) \cdot Z_{n,\infty}^{\frac{1}{\beta \cdot w}} + \frac{g_1 \cdot R_{n,1}(b_n)}{p_1 \cdot \tau} \right) \cdot e^{i\theta_n} \quad (5)$$

is thus evaluated without loss of precision or IEEE-754 floating-point jitter. The system thereby achieves 100% deterministic behavior across different processors, as the continuous phase $Z_{n,\infty}$ is seamlessly coupled to the discrete bit operations of the Hensel division via fixed-point arithmetic.

4.4 Epigenetic Serialization and Persistence Engine

To fulfill the requirement of long-term behavioral persistence, the system incorporates a binary serialization layer[cite: 9]. Upon graceful shutdown (triggered via system signals or user interaction), the complete vector of all 17 core registers $R_{\text{dna}} = (r_0, r_1, \dots, r_{16})$ is dumped directly into a persistent local file designated as `mycelium_dna.dat`[cite: 9].

Upon subsequent system initialization, this file is read back into memory to prime the core registers[cite: 9]. This mechanism bypasses traditional machine learning weight training (back-propagation), implementing an “epigenetic” memory where historical interactions continuously shape the evolutionary trajectory of the digital organism across operational lifetimes[cite: 9].

5 Experimental Evaluation and State-Space Dynamics

5.1 Experimental Setup and Stimulus Injection

To validate the real-time operational characteristics and state-space exploration capabilities of the Adelic-Irrational Mycelium Network, the 17-core architecture was deployed on an ultra-low-power edge microcontroller testbed[cite: 9]. The system was subjected to continuous real-time environmental data streams, specifically transient acoustic inputs and transducer voltage fluctuations captured via the streamlined AMDF pipeline[cite: 9]. Stimuli were digitized into paired vectors $S = (p, e)$, continuously driving the root node c_0 while secondary cores executed decentralized bit-diffusion loops concurrently[cite: 9].

5.2 Ergodic Exploration and Aperiodicity Verification

A critical vulnerability of conventional bit-level cellular automata and shift registers is their susceptibility to short limit cycles and periodic trapping states[cite: 9]. To quantify the efficacy of our irrational weighting scheme (W), we tracked the global system state $\bar{R}^{(t)} = \bigoplus_{i=0}^{16} r_i^{(t)}$ over 10^7 discrete execution cycles[cite: 9]:

- **Autocorrelation Analysis:** Empirical autocorrelation functions computed on the register traces confirmed a complete absence of secondary peaks or periodic recurrences, demonstrating true aperiodic behavior[cite: 9].
- **Phase Space Coverage:** The integration of fundamental irrational constants (ϕ, π, e) as core multipliers ensured an unweighted, dense coverage of the 64-bit integer state space without introducing floating-point computational bloat[cite: 9].

5.3 Stability and Channel Decoupling

We evaluated the stability of the adelic system model under abrupt, high-energy boundary perturbations (simulating severe sensor noise or physical shock to soft-substrate transducers)[cite: 9]:

- **Archimedean Macro-Stability:** The continuous phase space component $Z_{n,\infty}$ maintained smooth macroscopic trajectories, acting as an effective low-pass stabilizer for global pattern recognition[cite: 9].

- **Non-Archimedean Isolation:** The discrete p -adic prime channels Z_{n,p_i} successfully isolated localized high-frequency disturbances within their respective prime-indexed subspaces, preventing systemic data corruption and validating the theoretical suppression of cross-talk[cite: 9].

5.4 Epigenetic Persistence and Adaptation Benchmarks

Long-term adaptation performance was benchmarked by evaluating the recovery and alignment speed of the network following re-initialization from the `mycelium_dna.dat` serialized state file[cite: 9]. Compared to a cold-booted network, pre-primed instances demonstrated an immediate convergence toward user-specific stimulus response patterns, bypassing the typical transient warm-up phase and verifying the viability of historical epigenetic memory at the edge[cite: 9].

6 Practical Use Cases and Bio-Sensor Applications

6.1 Decentralized Environmental and Agricultural Monitoring

In subterranean or dense canopy environments where continuous cloud connectivity and fixed power grids are absent, the Mycelium Network offers a self-sustaining computational paradigm[cite: 9]. Sensor nodes embedded in organic soil substrates utilize local AMDF signal acquisition to process acoustic, hydraulic, or chemical fluctuations[cite: 9]. The decentralized p -adic diffusion across the 17 cores enables local anomaly detection and pattern evolution without high-throughput wireless data transmission or high-power central gateways[cite: 9].

6.2 Soft Robotics and Biomimetic Tactile Skins

Soft robotic actuators require compliant, distributed sensing to process non-linear structural deformations[cite: 9]. By mapping multi-point pressure and strain transducers directly into the root register r_0 , dynamic strain waves propagate through the core ring topology[cite: 9]. The resulting folded state $R^{(t)}$ generates real-time control voltage parameters for fluidic or electro-active actuators, enabling decentralized reflex responses analogous to peripheral nervous systems in biological organisms[cite: 9].

6.3 Edge-AI in Zero-Cloud and Ultra-Low-Power Environments

Traditional edge machine learning models suffer from static parameterization and high energy demands during continuous fine-tuning[cite: 9]. The epigenetic persistence engine (`mycelium_dna.dat`) enables edge devices to adapt to local environmental signatures across operational lifecycles using basic 64-bit integer operations ($\mathbb{U}_{64}$)[cite: 9]. This provides a long-term, self-evolving state engine suited for energy-harvesting, zero-cloud computing systems[cite: 9].

7 Conclusion

This paper presented the Adelic-Irrational Mycelium Network, a biomimetic 17-core architecture bridging higher number theory with low-level bitwise processing[cite: 9]. By embedding global dynamics within an adelic ring structure, applying irrational constants (ϕ, π, e) for ergodic state-space exploration, and incorporating AMDF-based signal acquisition, the architecture achieves robust, non-periodic signal diffusion without continuous floating-point computations[cite: 9]. The integrated epigenetic persistence layer guarantees long-term historical adaptation without backpropagation overhead[cite: 9]. Future work will investigate multi-ring hierarchical topologies and physical implementations on neuromorphic micro-architectures[cite: 9].